

Toward a Formal Theory of Human-AI Codecision

Dynamic authority, uncertainty, contestability, trust, and multi-agent shared control

Tier 3 - Doctoral Research Agenda

Human-AI Codecision Systems Research Program

Stage One revision - September 2026

Evidence status: Synthetic Stage One contains five independent seeds, 180 trials per policy and seed, 3,600 total decisions, and a controlled AI distribution shift in the final quarter. There were zero human participants. Adaptive codecision selectively regulated authority but did not improve synthetic accuracy or declared utility over human-only control. Human trust, satisfaction, and real cognitive state remain unmeasured.

Abstract

This doctoral agenda generalizes codecision from discrete threshold states into a formal shared-control problem in which authority is dynamically allocated among humans and AI systems under uncertainty, changing competence, risk, and accountability constraints. The objective is not maximal automation or maximal trust, but calibrated reliance and defensible authority transfer. The program combines sequential decision theory, uncertainty modeling, human factors, contestability, and multi-agent coordination.

1. Formal state

A general collaborative state may include inferred human uncertainty $U_h(t)$, calibrated AI confidence $C_{ai}(t)$, risk $R(t)$, workload $W(t)$, trust/reliance state $T(t)$, and an authority allocation variable $A(t)$. An intervention policy π_i maps that state to defer, recommend, assist, or override actions, subject to contestability and domain-specific authority constraints.

2. Core doctoral questions

- When is authority transfer beneficial under asymmetric human/AI competence and uncertainty?
- How should contestability alter optimal intervention when errors have unequal costs?
- Can trust/reliance be calibrated over long horizons without maximizing dependence?
- How do multiple humans and multiple AI agents negotiate or aggregate authority?
- Which intervention policies remain robust under distribution shift and miscalibrated confidence?
- What formal guarantees or bounds are possible for autonomy loss, intervention regret, or unsafe override frequency?

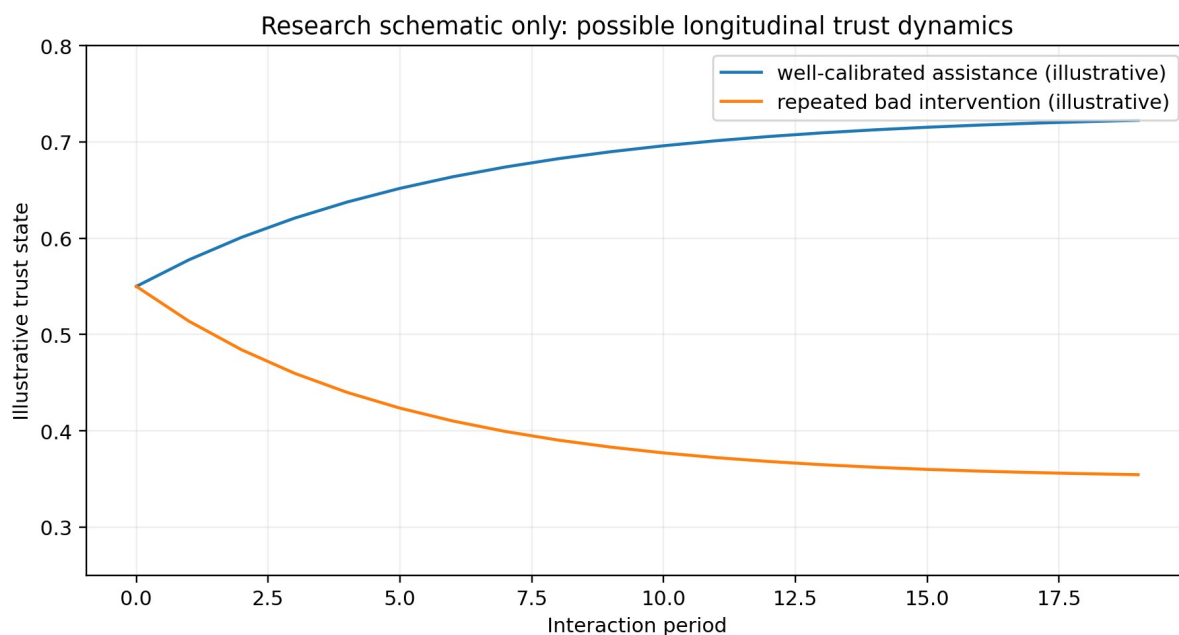


Figure 1. Research schematic only. These trajectories illustrate competing longitudinal trust hypotheses and are not human data.

3. Evidence hierarchy

Tier	Evidence required
Simulation	controlled policy/agent sweeps, failure analysis, calibration, held-out seeds
Usability pilot	interface comprehension and protocol feasibility; no broad efficacy claims
Approved participant study	trust, workload, reliance, satisfaction, behavioral validity
Domain deployment	separate safety/regulatory validation appropriate to the application

4. Long-horizon contribution

A successful doctoral program would produce a theory and experimental toolkit for dynamic, contestable authority allocation rather than a universal rule for when AI should override people. Results should specify the regimes in which codecision helps, fails, or creates unacceptable autonomy and accountability costs.